Relining and Rebasing

Relining is defined as, "A procedure to resurface the tissue surface of the denture with new base material to make the denture fit more accurately" - GPT.

Rebasing "A process of refitting a denture by the replacement of the denture base material" - GPT.

Rebasing is similar to relining except that there is extensive replacement of the denture base material.

Contraindications for Relining and Rebasing:

1. When the residual ridge has resorbed excessively.
2. Abused soft tissues due to an ill-fitting denture.
3. Temporo-mandibular joint problems.
4. Patient dissatisfied with the appearance of the existing dentures.
5. Unsatisfactory jaw relationships in the denture.
6. Dentures causing major speech problems.
7. Severe osseous undercuts.

A number of changes can occur in the tissues that support complete dentures. They are more common under *mandibular* than under *maxillary* dentures, but they may be encountered under either, particularly when an upper denture is opposed by the natural dentition. The magnitude of the changes is what determines the nature of the resurfacing or refitting prescribed. If a new thin layer of resin is added to the denture base, the resurfacing is called a reline. If more material is added when extensive refitting and polymerized material is necessary, this is called a rebase.

Dr. Wasmaa Sadik

Indication for Relining and Rebasing:

1. Loss of retention and stability
2. Loss of vertical dimension of occlusion
3. Loss of support for facial tissues.
4. Due to financial reason, the patient can't afford a new denture, and no contra indications apply.
5. Medical and social reasons prevent the patient from attending the required number of visits for a new complete denture and no contra indications apply.

Relining Procedures

• **Clinical procedures**:

1. Static methods:

A• Open-mouth technique.

B• Closed-mouth technique.

2. Functional methods
3. Chair-side technique

• **Laboratory procedures:**

• Articulator method

• Jig method

• Flask method

Basic laboratory procedure for Relining:

1. Once the impression is received, a cast is poured immediately, that is an impression taken using the denture as a tray.

Dr. Wasmaa Sadik

2. The denture and the cast are not separated, but any excess impression on the teeth or facial surfaces of the base is removed, then the denture flasked in the usual manner.

3. Wax elimination by heating in hot water for 5 minutes, then separated and all impression materials cleaned from the cast and the denture base.

4. Painting the cast with a separating medium.

5. Paint the surface of the denture with cotton pellet moistened with monomer.

6. Mix the acrylic resin and place it in the flask.

7. Curing for 9 hours at 165 F.

8. after curing, deflask, trim the excess acrylic resin, smooth, polish.

9. The maxillary and mandibular dentures remounted on articulator for correction of occlusion, then the denture is ready for insertion in patient's mouth.

Laboratory Procedures for Rebasing:

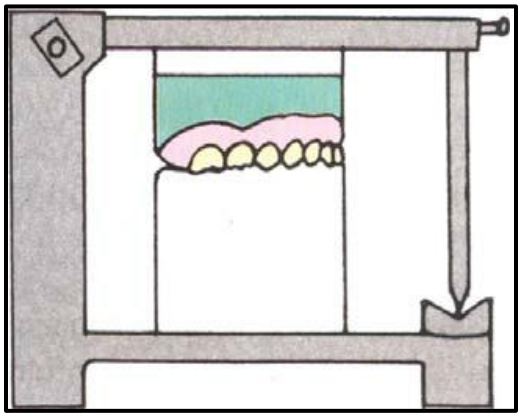
The laboratory procedures used for rebasing are the same as the ones used for relining. They include articulator method, flasking method and jig method. Irrespective of the methods used, rebasing differs from relining only in denture trimming prior to wax-up.

Denture Trimming Prior to Wax-up

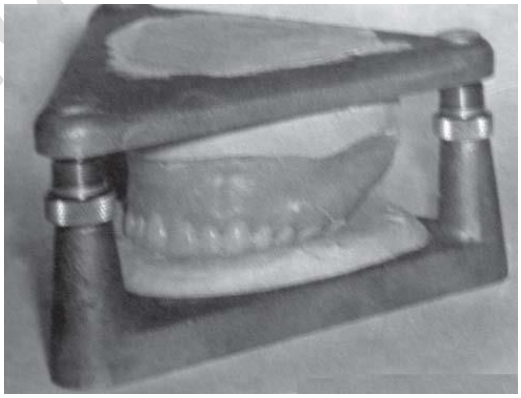
This is the only step where rebasing differs from relining. After articulating or flasking the cast (depending of the method) the denture is removed from the cast and the entire denture base is trimmed leaving just 2 mm of acrylic around the porcelain teeth. The acrylic is retained to preserve the positions of the denture

Dr. Wasmaa Sadik

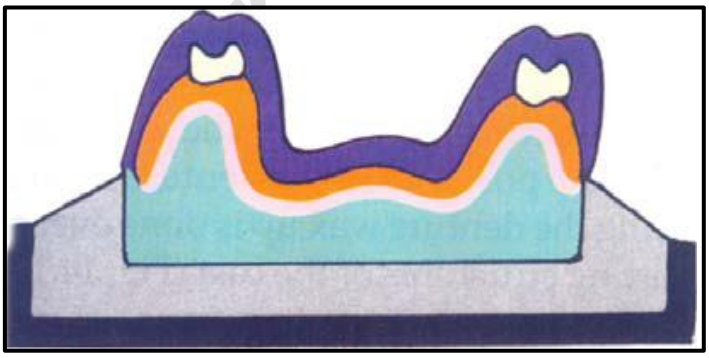
teeth. After trimming the denture wax up is done over all the supporting structures of the cast.



(fig. 1 Articulating the denture (with impression) and cast against a plaster)



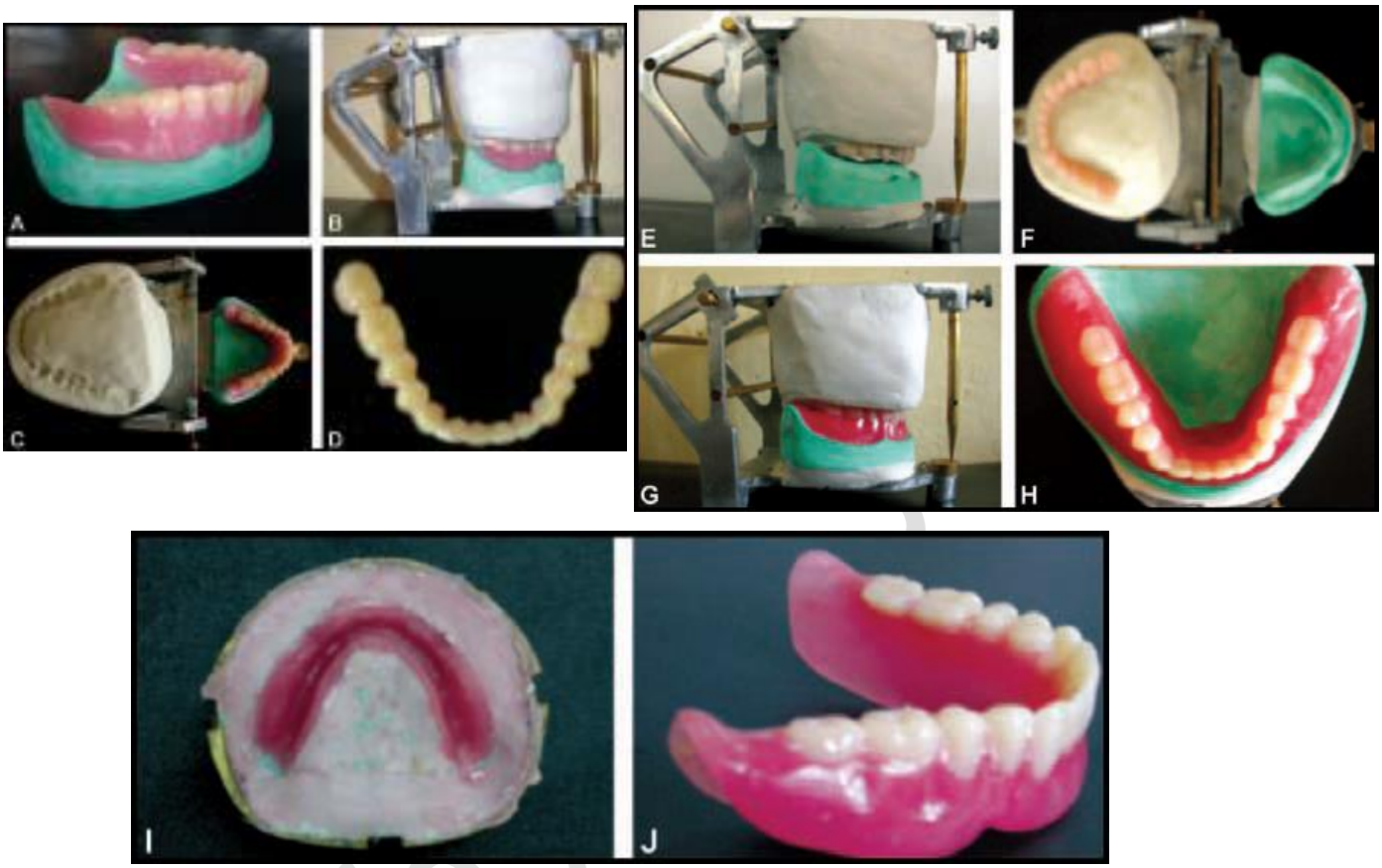
(fig.2 Hooper's duplicator, Jig method).



(fig.3 Flask method).

Dr. Wasmaa Sadik

(Fig. 4) Relining of mandibular complete denture (A) Denture flanges and tissue surface reduced by 2mm with stops in the canine and molar region (B) Border molding with low fusing stick compound (C) Final impression (D) The cast along with denture invested in the dental flask (E) Removal of low fusing compound and impression material (F) Acrylic packed on the tissue surface of the denture (G), (H), (I) Relined denture



(Fig. 5) Rebasing of mandibular denture. (A) Porosities in the mandibular denture (B),(C) Index of occlusal surfaces of the mandibular denture (D) String of teeth cut away from the denture base (E),(F) String of teeth placed in the index (G),(H) Waxup of new denture base (I)(J). Rebased the mandibular denture.